Siu Lun Chau

Assistant Professor @ College of Computing & Data Science, Nanyang Technological University
Principal Investigator @ Epistemic Intelligence & Computation Lab

Research Interests

My research focuses on building safe and reliable AI by advancing the principles necessary to develop epistemically intelligent systems – machine learning systems that can recognise, reason about, and communicate the limits of their knowledge. To achieve this, my research spans foundational theory to translational applications of imprecise probabilities as the main pillar for trustworthy AI.

Work Experiences

2025–Present	**Assistant Professor**, College of Computing & Data Science, NTU Singapore
2023–2025	**Postdoctoral Researcher**, CISPA Helmholtz Center for Information Security, Germany. Supervisor: Dr. Krikamol Muandet
2023–2024	**Visiting Researcher**, Australian Institute for Machine Learning
2023–2023	**Research Assistant**, CISPA Helmholtz Center for Information Security, Germany
2022–2022	**Applied Science Intern**, Amazon UK
2021–2022	**Research Intern**, Max Planck Institute for Intelligent Systems. Supervisor: Dr. Krikamol Muandet

Education

2018–2023	**DPhil in Statistics**, University of Oxford. EPSRC & MRC Studentship. Supervisor: Prof. Dino Sejdinovic.
2014–2018	**MMath in Mathematics and Statistics**, University of Oxford. Ranked 1st in 3rd year. Master thesis supervisors: Prof. Mihaela van der Schaar and Prof. Geoff Nicholls.

Awards & Recognition

- ICML Silver Reviewer Award.
- Global Research Excellence Award for Travel (Faculty award) 2025. NTU.
- First Prize, IJAR Young Researcher Award 2025.
- ICML 2026 Spotlight • UAI 2025 Oral • ICML 2024 Spotlight • AAAI 2024 Oral • NeurIPS 2023 Spotlight.
- EPSRC & MRC Doctoral Studentship. 2018–2023.
- Department Prize for FHS Mathematics and Statistics Part B, 2017. University of Oxford.

Publications

Total citations: **514** • h-index: **14** • i10-index: **16**.

Source: Google Scholar, as of July 2026.

Conference Papers

C24. Zhang, Y., Muandet, K., Sejdinovic, D., Fong, E., **Chau, S.L.** "Instrumental and Proximal Causal Inference with Gaussian Processes." *The Conference on Uncertainty in Artificial Intelligence*, 2026. [pdf]

C23. Yang, A., Muandet, K., Caprio, M., **Chau, S.L.**, Adachi, M. "Verbalizing LLM's Higher-order Uncertainty via Imprecise Probabilities." *The Conference on Uncertainty in Artificial Intelligence*, 2026. [pdf]

C22. Wang, K., Wang, Y., Cuzzolin, F., Moens, D., Hallez, H., **Chau, S.L**. "Set-based vs Distribution-based Representations of Epistemic Uncertainty: A Comparative Study." *The Conference on Uncertainty in Artificial Intelligence*, 2026. [pdf]

C21. Wang, K., Faza, G.A., Cuzzolin, F., **Chau, S.L.**, Moens, D., Hallez, H. "Learning Credal Ensembles via Distributionally Robust Optimization." *International Conference on Machine Learning*, 2026. [pdf] **(spotlight)**

C20. Vo, K.Q., **Chau, S.L.**, Kato, M., Wang, Y., Muandet, K. "Explanation Design in Strategic Learning: Sufficient Explanations that Induce Non-harmful Responses." *The 29th International Conference on Artificial Intelligence and Statistics*, 2026. [pdf] [code]

C19. Vo, K.Q., Reddy, A.G., Rodemann, J., **Chau, S.L.**, Muandet, K. "Off-Policy Evaluation with Strategic Agents via Local Disclosure." *International Conference on Machine Learning*, 2026.

C18. Mohammadi, M., Muandet, K., Tiddi, I., Ten Teije, A., **Chau, S.L**. "Exact shapley attributions in quadratic-time for fanova gaussian processes." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. [pdf]

C17. Gonzalez-Garcia, X., **Chau, S.L.**, Rodemann, J., Caprio, M., Muandet, K., Bustince, H., Destercke, S., Hüllermeier, E., Sale, Y. "Quantification of Credal Uncertainty: A Distance-Based Approach." *The Conference on Uncertainty in Artificial Intelligence*, 2026. [pdf]

C16. Caprio, M., Papagiannouli, K., **Chau, S.L.**, Mukherjee, S. "Robust Predictive Uncertainty and Double Descent in Contaminated Bayesian Random Features." *The Conference on Uncertainty in Artificial Intelligence*, 2026. [pdf]

C15. Singh, A., **Chau, S.L.**, Muandet, K. "Truthful Elicitation of Imprecise Forecasts." *Conference on Uncertainty in Artificial Intelligence*, 2025. [pdf] [code] **(Oral)**

C14. Naslidnyk, M., **Chau, S.L.**, Briol, F.-X., Muandet, K. "Kernel Quantile Embeddings and Associated Probability Metrics." *International Conference on Machine Learning*, 2025. [pdf]

C13. **Chau, S.L.**, Caprio, M., Muandet, K. "Integral Imprecise Probability Metrics." *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. [pdf]

C12. **Chau, S.L.**, Schrab, A., Gretton, A., Sejdinovic, D., Muandet, K. "Credal Two-Sample Tests of Epistemic Uncertainty." *International Conference on Artificial Intelligence and Statistics*, 2025. [pdf] [code] [vid

C11. Vo, K.Q., Aadil, M., **Chau, S.L.**, Muandet, K. "Causal strategic learning with competitive selection." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024. [pdf] [code] **(Oral)**

C10. Singh, A., **Chau, S.L.**, Bouabid, S., Muandet, K. "Domain generalisation via imprecise learning." *Proceedings of the 41st International Conference on Machine Learning*, 2024. [pdf] [code] **(Spotlight)**

C9. Adachi, M., Planden, B., Howey, D., Osborne, M.A., Orbell, S., Ares, N., Muandet, K., **Chau, S.L**. "Looping in the Human: Collaborative and Explainable Bayesian Optimization." *International Conference on Artificial Intelligence and Statistics*, 2024. [pdf] [code]

C8. **Chau, S.L.**, Muandet, K., Sejdinovic, D. "Explaining the uncertain: Stochastic shapley values for gaussian process models." *Advances in Neural Information Processing Systems*, 2023. [pdf] [code] [video] **(Spotlight)**

C7. Hu, R., **Chau, S.L.**, Sejdinovic, D., Glaunès, J. "Giga-scale kernel matrix-vector multiplication on GPU." *Advances in Neural Information Processing Systems*, 2022. [pdf] [code]

C6. Hu*, R., **Chau*, S.L.**, Ferrando Huertas, J., Sejdinovic, D. "Explaining preferences with shapley values." *Advances in Neural Information Processing Systems*, 2022. [pdf] [code] [video]

C5. **Chau, S.L.**, Cucuringu, M., Sejdinovic, D. "Spectral ranking with covariates." *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, 2022. [pdf] [code]

- **C4.** **Chau, S.L.**, Hu, R., Gonzalez, J., Sejdinovic, D. “RKHS-SHAP: Shapley values for kernel methods.” *Advances in neural information processing systems*, 2022. [pdf] [code] [video]
- **C3.** **Chau, S.L.**, Gonzalez, J., Sejdinovic, D. “Learning inconsistent preferences with gaussian processes.” *International Conference on Artificial Intelligence and Statistics*, 2022. [pdf] [video]
- **C2.** **Chau*, S.L.**, Bouabid*, S., Sejdinovic, D. “Deconditional downscaling with gaussian processes.” *Advances in Neural Information Processing Systems*, 2021. [pdf] [code] [video]
- **C1.** **Chau*, S.L.**, Ton*, J.-F., González, J., Teh, Y., Sejdinovic, D. “Bayesimp: Uncertainty quantification for causal data fusion.” *Advances in Neural Information Processing Systems*, 2021. [pdf] [video]

Workshop & Non-Archival Papers

- **WS5.** Wang, K., Wang, Y., Cuzzolin, F., Moens, D., Hallez, H., **Chau, S.L**. “Set-based vs Distribution-based Representations of Epistemic Uncertainty: A Comparative Study.” *ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML)*, 2026.
- **WS4.** Sin, J.W., Segura, D.M., Ranković, B., **Chau, S.L.**, Burwood, R.P., Püntener, K., Bigler, R., Schwaller, P. “Dynamic Large Language Model Representations for Multi-Objective Chemical Reaction Optimisation.” *ICML 2026 Workshop on AI for Science (AI4Science)*, 2026.
- **WS3.** Chong, C.H., Wang, K., Caprio, M., **Chau, S.L**. “How do Consonance Constructions Shape Conformal Credal Sets?.” *ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML)*, 2026.
- **WS2.** Singh, A., Rodemann, J., Verma, R., **Chau, S.L.**, Muandet, K. “Incentive Aware AI Regulation.” *EurIPS PAIG Workshop (Poster)*, 2025.
- **WS1.** **Chau, S.L.**, Schrab, A., Gretton, A., Sejdinovic, D., Muandet, K. “Credal Two-Sample Tests of Epistemic Uncertainty.” *Poster presentation at 14th International Symposium on Imprecise Probabilities: Theories and Applications (ISIPTA)*, 2025.

Journal Papers

- **J3.** Sin, J.W., **Chau, S.L.**, Burwood, R.P., Püntener, K., Bigler, R., Schwaller, P. “Highly parallel optimisation of chemical reactions through automation and machine intelligence.” *Nature Communications*, 2025. [pdf] [code]
- **J2.** Föll, S., Dubatovka, A., Ernst, E., **Chau, S.L.**, Maritsch, M., Okanovic, P., Thaeter, G., Buhmann, J.M., Wortmann, F., Muandet, K. “Gated Domain Units for Multi-source Domain Generalization.” *Transactions on Machine Learning Research*, 2023. [pdf] [code]
- **J1.** Pu, X., **Chau, S.L.**, Dong, X., Sejdinovic, D. “Kernel-based graph learning from smooth signals: A functional viewpoint.” *IEEE Transactions on Signal and Information Processing over Networks*, 2021. [pdf]

Preprints

- **W7.** Singh, A., Rodemann, J., Verma, R., **Chau, S.L.**, Muandet, K. “Incentive Aware AI Regulations: A Credal Characterisation.” *arXiv:2603.05175*. [pdf] [code]
- **W6.** Moskvichev, P., **Chau, S.L.**, Sejdinovic, D. “Measuring Differences between Conditional Distributions using Kernel Embeddings.” *arXiv:2605.02260*.
- **W5.** Mohammadi, M., Reznikov, G., Sinitcyn, P., Muandet, K., **Chau, S.L**. “QuadraSHAP: Stable and Scalable Shapley Values for Product Games via Gauss–Legendre Quadrature.” *arXiv:2605.05870*. [pdf]
- **W4.** Mohammadi*, M., **Chau*, S.L.**, Muandet, K. “Amortized Linear-time Exact Shapley Value for Product-Kernel Methods.” *arXiv:2505.16516*. [pdf]
- **W3.** **Chau, S.L.**, Zargarbashi, S.H., Sale, Y., Caprio, M. “Quantifying Epistemic Predictive Uncertainty in Conformal Prediction.” *arXiv:2602.01667*. [pdf] [code]
- **W2.** Caprio, M., **Chau, S.L.**, Muandet, K. “When do credal sets stabilize? fixed-point theorems for credal set updates.” *arXiv:2510.04769*. [pdf]
- **W1.** Adachi, M., **Chau, S.L.**, Xu, W., Singh, A., Osborne, M.A., Muandet, K. “Bayesian optimization

for building social-influence-free consensus." *arXiv:2502.07166*. [pdf]

Theses

T1. "Towards trustworthy machine learning with kernels." *University of Oxford*, 2023. [pdf]

Invited Presentations

Guest Lectures

2026 SIPTA (Society of Imprecise Probability: Theory and Application) Summer School

Invited Talks

University of Oxford (2026, 2022); University of Manchester (2026); LMU Munich (2025, 2024); University of Hong Kong (2025); Chinese University of Hong Kong (2025); City University of Hong Kong (2025); A*STAR (2025); DFKI (2025); Toyota Motor Corporation (2025); NUS (2025); Amazon Web Services Berlin (2025); UCL Statistical Science (2024); Gatsby Computational Neuroscience Unit, UCL (2024, 2022); University of Adelaide (2024); NTU (2024); ETH AI Center (2023); ETH D-MTEC (2023); Australian Data Science Network (2023); Data61, Melbourne (2023); Australian National University (2023); University of Melbourne (2023); AIML Adelaide (2023); Oxford-Man Institute (2023); Oxford Strategy Group Digital (2023); CISPA (2023); ELISE Theory Workshop, Eurecom (2022); ECML-PKDD (2022); Alan Turing Institute (2022); Imperial & Oxford STATML Seminar (2022); Warwick ML Group (2021).

Reviewing Service

Conference

NeurIPS (2025, 2024, 2023, 2022, 2021); ICML (2025, 2024); AAAI (2025); AISTATS (2025, 2024, 2022); STAI-X (2026); UAI (2025); ECML-PKDD (2022).

Journal

Journal of the American Statistical Association (JASA); Pattern Analysis and Machine Intelligence (TPAMI); Transactions on Machine Learning Research (TMLR); International Journal of Approximate Reasoning (IJAR); Statistics and Computing; Fuzzy Sets and Systems.

Workshop Organisation & Professional Memberships

Workshops

Organiser	Second Workshop on Epistemic Intelligence in Machine Learning, ICML 2026
Organiser	First Workshop on Epistemic Intelligence in Machine Learning, EurIPS 2025

Memberships

2026–Present	Member, Institute of Mathematical Statistics
2025–Present	Member, Association for the Advancement of Artificial Intelligence (AAAI)
2025–Present	Fellow, Royal Statistical Society
2024–Present	Member, The Society for Imprecise Probabilities: Theory and Applications (SIPTA)

Research Funding

2025–2028	Foundations and Applications of Epistemic Machine Learning (PI). S$150,000, NTU Start-Up Grant.
2025–2025	Sponsorship for EIML@EurIPS 2025 (Applicant). GBP 4,000, ELLIS Unit Manchester Event Support Fund.
2026–2026	Sponsorship for EIML@ICML 2026 (Applicant). GBP 4,000, ELLIS Unit Manchester Event Support Fund.
2026–2027	Funding for Visiting Student (Applicant). USD 3,900, IEEE CIS Education Graduate Student Grants.

Teaching

SC1004	Linear Algebra for Computing — Tutorial (Sem 1) & Lecture (Sem 2), AY2025–26
SC3021	Foundations of Data Science — Tutorial (Sem 2), AY2025–26

Academic Supervision

Postdoctoral Fellows

- **Kaizheng Wang** (Nanyang Technological University Singapore, Postdoctoral Research Fellow, 2026–Present). Project: *Epistemic Intelligence in Machine Learning*.

PhD Students

- **Chun Ho Chong** (Nanyang Technological University Singapore, PhD Supervisor, Jan 2026–Present). Project: *Uncertainty Explanations*.
- **Joshua Adrian Cahyono** (Nanyang Technological University Singapore, PhD Supervisor, Jan 2026–Present). Project: *Proper Uncertainty Management for AI Safety*.
- **Yuqi Zhang** (The University of Hong Kong, PhD Co-Supervisor, Jan 2026–Present). Project: *Uncertainty-aware Causal Inference (main supervisor: Prof. Edwin Fong, HKU)*.

Undergraduate & Master Students

- **Justin Tang Jing Han** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Learning Rashomon Sets of Representations*.
- **Hng Cherng Khai** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Exploring the Evaluation of Uncertainty-aware Classifiers beyond Image Tasks*.
- **Wang Shi Ying** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Machine Learning meets Econometrics: Difference-in-Differences*.
- **Solis Aaron Mari Santos** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Machine Learning meets Econometrics: Regression Discontinuity Design*.
- **Ng Zheng Da** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Epistemic Uncertainty Quantification for Rating Data*.
- **Chew Ming Hong Gordon** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Epistemic Uncertainty Quantification in Pairwise Comparison Preference Data*.
- **Tan Wee Choon Waylen** (Nanyang Technological University Singapore, Final Year Project, May 2026–Present). Project: *Stochastic Shapley values for Probabilistic Models*.
- **Xiaotao Liu** (Nanyang Technological University Singapore, Undergraduate Research Programme Supervisor, June 2025–Present). Project: *Deep Ensemble surrogates for Bayesian Optimisation*.
- **Swathi Suhas** (Saarland University, Master Student, Jan 2024–Jan 2025). Project: *Explaining Kernel-based Outlier Detectors with Shapley Values*.
- **Oscar Yung** (University of Oxford, Master Student, Sep 2021–Mar 2022). Project: *MMD Two-Sample Testing in Regression Discontinuity Design*.
- **Samuel Weinman** (University of Oxford, Master Student, June 2020–Sep 2020). Project: *On Price-Volume Interplay in Equity Markets via Dimensionality Reduction and Kernel Methods*.

PhD Thesis Examinations

- **Wang Ziyan**, Ph.D., College of Computing & Data Science, NTU, *Bridging Semantic and Collaborative Signals for LLM-based Recommendation*, 2026.
- **Li Haoxin**, Ph.D., College of Computing & Data Science, NTU, *Mitigating Shortcut Learning and Improving Out-of-Distribution Generalization for Vision and Vision-Language Models*, 2026.
- **Deng Zichao**, Ph.D., College of Computing & Data Science, NTU, *Towards Trustworthy Graph Learning: Robustness, Privacy, and Fairness*, 2026.

- **Dong Junhao**, Ph.D., College of Computing & Data Science, NTU, *Towards Generalizing Adversarial Robustness in Deep Learning*, 2026.